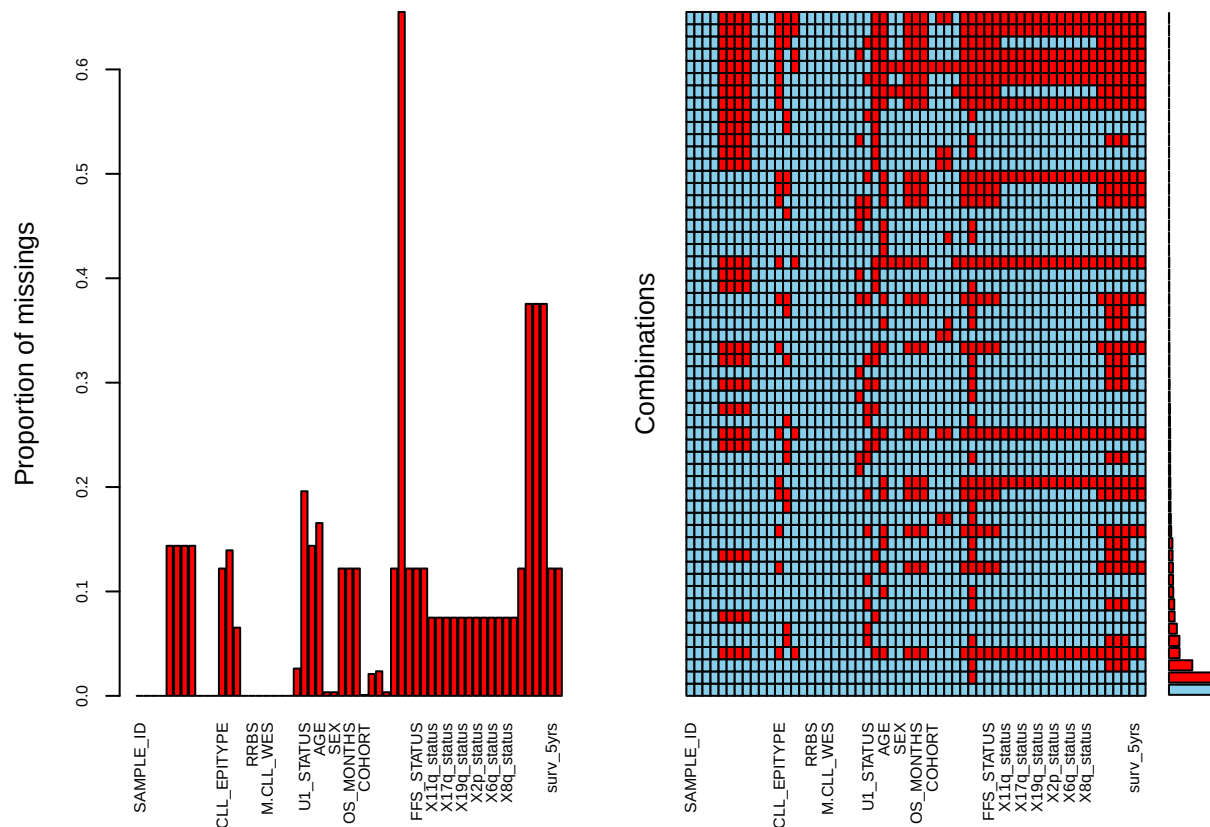


visualize_missing_data

```
data <- read.table("/Users/sorenbasnet/Documents/Github/STAT_443_FINAL_PROJECT/data/importance_score/pa
library(VIM)
```

```
## Loading required package: colorspace
## Loading required package: grid
## VIM is ready to use.
## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues
##
## Attaching package: 'VIM'
## The following object is masked from 'package:datasets':
##
##     sleep
par(cex=0.7)
aggr(data)
```

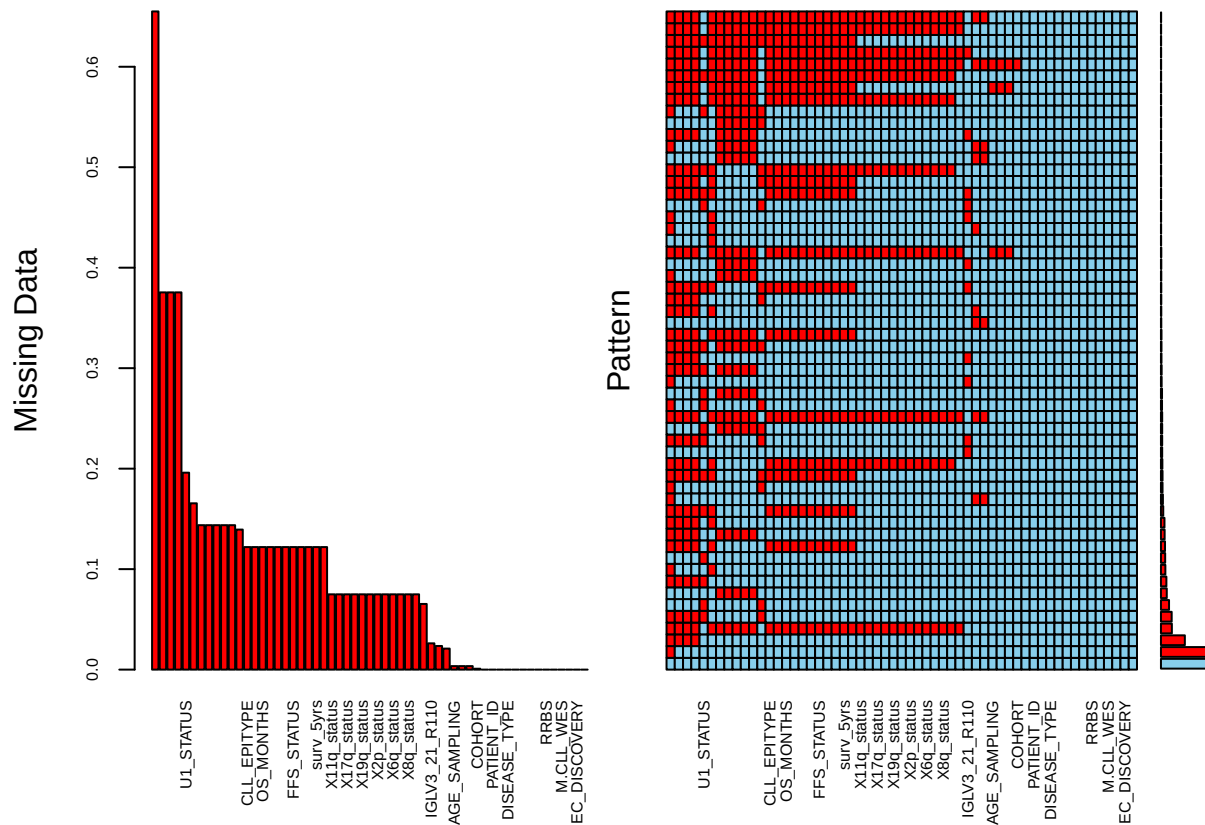


```
#par(cex = 0.8) # slightly bigger text
#par(mar = c(10, 4, 4, 2) + 0.1, cex = 0.8)
```

```
# 2. Set margins: c(bottom, left, top, right)
par(mar = c(12, 5, 2, 2), cex = 0.8)
#aggr(data,
#      col = c("skyblue", "red"),      # blue = observed, red = missing
#      numbers = TRUE,                  # show counts
#      sortVars = TRUE,                 # sort by missingness
#      labels = names(data),            # variable names
#      cex.axis = 0.7,                  # axis text size
#      gap = 3,                         # spacing between bars
#      ylab = c("Missing data", "Pattern"))
```

```
aggr(data,
      col = c("skyblue", "red"),
      numbers = TRUE,
      sortVars = TRUE,
      labels = names(data),
      cex.axis = 0.6, # Smaller text helps labels fit
      gap = 2,        # Reduced gap can help squeeze more in
      ylab = c("Missing Data", "Pattern"))
```

```
## Warning in plot.aggr(res, ...): not enough vertical space to display
## frequencies (too many combinations)
```



```
##
## Variables sorted by number of missings:
##      Variable      Count
```

```

##      TREATMENT_AFTER_SAMPLING 0.6550522648
##      TREATMENT_STATUS_AT_SAMPLING 0.3754355401
##      RAI_AT_SAMPLING 0.3754355401
##      FIRST_TREATMENT_AFTER_SAMPLING 0.3754355401
##      U1_STATUS 0.1959930314
##      AGE 0.1655052265
##      NORMAL_MEAN_COVERAGE 0.1437282230
##      TUMOR_MEAN_COVERAGE 0.1437282230
##      TUMOR_SAMPLE_PURITY 0.1437282230
##      TUMOR_SAMPLE_PLOIDY 0.1437282230
##      TMB_NONSYNONYMOUS 0.1437282230
##      TUMOR_MOLECULAR_SUBTYPE 0.1393728223
##      CLL_EPITYPE 0.1219512195
##      OS_STATUS 0.1219512195
##      OS_MONTHS 0.1219512195
##      DEATH_DAYS 0.1219512195
##      PRIOR_TREATMENT_CATEGORY 0.1219512195
##      EXPRESSION_CLUSTER 0.1219512195
##      FFS_STATUS 0.1219512195
##      FFS_MONTHS 0.1219512195
##      RAI_AT_DIAGNOSIS 0.1219512195
##      surv_5yrs 0.1219512195
##      death_5yr 0.1219512195
##      X11q_status 0.0749128920
##      X17p_status 0.0749128920
##      X17q_status 0.0749128920
##      X18p_status 0.0749128920
##      X19q_status 0.0749128920
##      X20p_status 0.0749128920
##      X2p_status 0.0749128920
##      X4p_status 0.0749128920
##      X6q_status 0.0749128920
##      X8p_status 0.0749128920
##      X8q_status 0.0749128920
##      tri12_status 0.0749128920
##      SEQUENCING_PLATFORM 0.0653310105
##      IGLV3_21_R110 0.0261324042
##      IGHV_IDENTITY_PERCENTAGE 0.0235191638
##      IGHV_MUTATION_STATUS 0.0209059233
##      AGE_SAMPLING 0.0034843206
##      SEX 0.0034843206
##      TREATMENT_STATUS 0.0034843206
##      COHORT 0.0008710801
##      SAMPLE_ID 0.0000000000
##      PATIENT_ID 0.0000000000
##      ONCOTREE_CODE 0.0000000000
##      DISEASE_TYPE 0.0000000000
##      CANCER_TYPE 0.0000000000
##      CANCER_TYPE_DETAILED 0.0000000000
##      SOMATIC_STATUS 0.0000000000
##      RNA_SEQUENCING 0.0000000000
##      RRBS 0.0000000000
##      ARRAY450K 0.0000000000
##      M.CLL_WES 0.0000000000

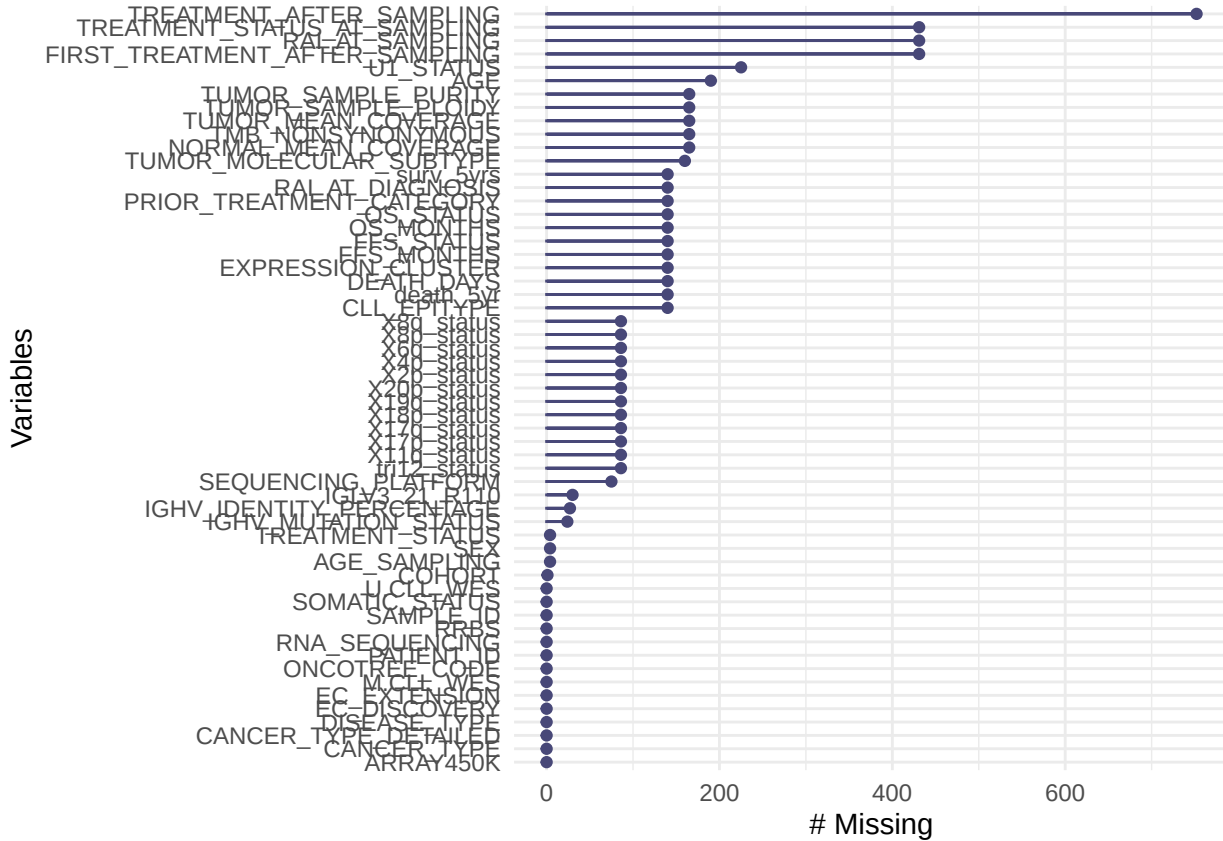
```

```
##          U CLL_WES  0.0000000000
##          EC_DISCOVERY 0.0000000000
##          EC_EXTENSION 0.0000000000
```

Various methods we explored to show the all column names in the x-axis.

```
library(naniar)
```

```
gg_miss_var(data)
```



We can see that status of treatment after sampling has the most number of missing values followed by variable having to do with treatment status and rai stage sampling which could've been done during sampling.

Imputation of such variables is not recommended because some patients might've opted for new medical advice or might've decided to not to proceed with recommended procedures.